

NMEA Standard Sentences

Here are the NMEA standard sentences we know about:

AAM - Waypoint Arrival Alarm

This sentence is generated by some units to indicate the status of arrival (entering the arrival circle, or passing the perpendicular of the course line) at the destination waypoint.

1 2 3 4 5 6

| | | | | |

\$--AAM,A,A,x.x,N,c--c*hh<CR><LF>

Field Number:

1. Status, BOOLEAN, A = Arrival circle entered, V = not passed
2. Status, BOOLEAN, A = perpendicular passed at waypoint, V = not passed
3. Arrival circle radius
4. Units of radius, nautical miles
5. Waypoint ID
6. Checksum

Example: GPAAM,A,A,0.10,N,WPTNME*43

WPTNME is the waypoint name.

ALM - GPS Almanac Data

This sentence expresses orbital data for a specified GPS satellite.

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

| | | | | | | | | | | | | | | |

\$--ALM,x.x,x.x,xx,x.x,hh,hhhh,hh,hhhh,hhhh,hhhhhh,hhhhhh,hhhhhh,hhh,hhh*h
h<CR><LF>

Field Number:

1. Total number of messages
2. Sentence Number
3. Satellite PRN number (01 to 32)
4. GPS Week Number
5. SV health, bits 17-24 of each almanac page
6. Eccentricity
7. Almanac Reference Time
8. Inclination Angle

9. Rate of Right Ascension
10. Root of semi-major axis
11. Argument of perigee
12. Longitude of ascension node
13. Mean anomaly
14. F0 Clock Parameter
15. F1 Clock Parameter
16. Checksum

Fields 5 through 15 are dumped as raw hex.

Example:

\$GPALM,1,1,15,1159,00,441d,4e,16be,fd5e,a10c9f,4a2da4,686e81,58cbe1,0a4,00
1*5B

APA - Autopilot Sentence "A"

This sentence is sent by some GPS receivers to allow them to be used to control an autopilot unit. This sentence is commonly used by autopilots and contains navigation receiver warning flag status, cross-track-error, waypoint arrival status, initial bearing from origin waypoint to the destination, continuous bearing from present position to destination and recommended heading-to-steer to destination waypoint for the active navigation leg of the journey.

1 2 3 4 5 6 7 8 9 10 11

| | | | | | | | |

\$--APA,A,A,x.xx,L,N,A,A,xxx,M,c---c*hh<CR><LF>

Field Number:

1. Status V = Loran-C Blink or SNR warning V = general warning flag or other navigation systems when a reliable fix is not available
2. Status V = Loran-C Cycle Lock warning flag A = OK or not used
3. Cross Track Error Magnitude
4. Direction to steer, L or R
5. Cross Track Units (Nautical miles or kilometers)
6. Status A = Arrival Circle Entered
7. Status A = Perpendicular passed at waypoint
8. Bearing origin to destination
9. M = Magnetic, T = True
10. Destination Waypoint ID
11. checksum

Example: \$GPAPA,A,A,0.10,R,N,V,V,011,M,DEST,011,M*82

APB - Autopilot Sentence "B"

This is a fixed form of the APA sentence with some ambiguities removed.

Note: Some autopilots, Robertson in particular, misinterpret "bearing from origin to destination" as "bearing from present position to destination". This is likely due to the difference between the APB sentence and the APA sentence. For the APA sentence this would be the correct thing to do for the data in the same field. APA only differs from APB in this one field and APA leaves off the last two fields where this distinction is clearly spelled out. This will result in poor performance if the boat is sufficiently off-course that the two bearings are different. 13 15

1 2 3 4 5 6 7 8 9 10 11 12| 14|

| | | | | | | | | | | |

\$--APB,A,A,x.x,a,N,A,A,x.x,a,c--c,x.x,a,x.x,a*hh<CR><LF>

Field Number:

1. Status A = Data valid V = Loran-C Blink or SNR warning V = general warning flag or other navigation systems when a reliable fix is not available
2. Status V = Loran-C Cycle Lock warning flag A = OK or not used
3. Cross Track Error Magnitude
4. Direction to steer, L or R
5. Cross Track Units, N = Nautical Miles
6. Status A = Arrival Circle Entered
7. Status A = Perpendicular passed at waypoint
8. Bearing origin to destination
9. M = Magnetic, T = True
10. Destination Waypoint ID
11. Bearing, present position to Destination
12. M = Magnetic, T = True
13. Heading to steer to destination waypoint
14. M = Magnetic, T = True
15. Checksum

Example: \$GPAPB,A,A,0.10,R,N,V,V,011,M,DEST,011,M,011,M*82

BOD - Bearing - Waypoint to Waypoint

1 2 3 4 5 6 7

| | | | | |

\$--BOD,x.x,T,x.x,M,c--c,c--c*hh<CR><LF>

Field Number:

1. Bearing Degrees, True

2. T = True
3. Bearing Degrees, Magnetic
4. M = Magnetic
5. Destination Waypoint
6. origin Waypoint
7. Checksum

Example 1: \$GPBOD,099.3,T,105.6,M,POINTB*01

Waypoint ID: "POINTB" Bearing 99.3 True, 105.6 Magnetic This sentence is transmitted in the GOTO mode, without an active route on your GPS. WARNING: this is the bearing from the moment you press enter in the GOTO page to the destination waypoint and is NOT updated dynamically! To update the information, (current bearing to waypoint), you will have to press enter in the GOTO page again.

Example 2: \$GPBOD,097.0,T,103.2,M,POINTB,POINTA*52

This sentence is transmitted when a route is active. It contains the active leg information: origin waypoint "POINTA" and destination waypoint "POINTB", bearing between the two points 97.0 True, 103.2 Magnetic. It does NOT display the bearing from current location to destination waypoint! WARNING Again this information does not change until you are on the next leg of the route. (The bearing from POINTA to POINTB does not change during the time you are on this leg.)

This sentence has been replaced by BWW in NMEA 4.00 (and possibly earlier versions) [\[ANON\]](#).

BWC - Bearing & Distance to Waypoint - Great Circle

12

1	2	3 4	5 6	7 8	9 10	11	13

```
$--BWC,hhmmss.ss,llll.ll,a,yyyyy.yy,a,x.x,T,x.x,M,x.x,N,c--c*hh<CR><LF>
```

NMEA 2.3:

```
$--BWC,hhmmss.ss,llll.ll,a,yyyyy.yy,a,x.x,T,x.x,M,x.x,N,c--c,m*hh<CR><LF>
```

Field Number:

1. UTC Time of observation, hh is hours, mm is minutes, ss.ss is seconds
2. Waypoint Latitude
3. N = North, S = South
4. Waypoint Longitude
5. E = East, W = West
6. Bearing, degrees True
7. T = True
8. Bearing, degrees Magnetic
9. M = Magnetic

- Example 1: \$GPBWC,081837,,,,,T,,M,,N*13
Example 2:
GPBWC,220516,5130.02,N,00046.34,W,213.8,T,218.0,M,0004.6,N,EGLM*11

[illegible]

- Bearing calculated at the FROM waypoint.
- | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|
| | | | | | | |
- \$--BWW,x.x,T,x.x,M,c--c,c--c*hh<CR><LF>
- Field Number:

1. Bearing, degrees True
2. T = True
3. Bearing Degrees, Magnetic
4. M = Magnetic
5. TO Waypoint ID
6. FROM Waypoint ID
7. Checksum

DBK - Depth Below Keel

1 2 3 4 5 6 7

| || || ||

\$--DBK,x.x,f,x.x,M,x.x,F*hh<CR><LF>

Field Number:

1. Depth, feet
2. f = feet
3. Depth, meters
4. M = meters
5. Depth, Fathoms
6. F = Fathoms
7. Checksum

DBS - Depth Below Surface

1 2 3 4 5 6 7

| || || ||

\$--DBS,x.x,f,x.x,M,x.x,F*hh<CR><LF>

Field Number:

1. Depth, feet
2. f = feet
3. Depth, meters
4. M = meters
5. Depth, Fathoms
6. F = Fathoms
7. Checksum

DBT - Depth below transducer

1 2 3 4 5 6 7

| || || ||

\$--DBT,x.x,f,x.x,M,x.x,F*hh<CR><LF>

Field Number:

1. Water depth, feet
2. f = feet
3. Water depth, meters
4. M = meters
5. Water depth, Fathoms
6. F = Fathoms
7. Checksum

In real-world sensors, sometimes not all three conversions are reported. So you might see something like \$SDDBT,,f,22.5,M,,F*cs

Example: \$SDDBT,7.8,f,2.4,M,1.3,F*0D

DCN - Decca Position

										11	13		16	
1	2	3	4	5	6	7	8	9	10	12	14	15	17	

\$--DCN,xx,cc,x.x,A,cc,x.x,A,cc,x.x,A,A,A,A,x.x,N,x*hh<CR><LF>

Field Number:

1. Decca chain identifier
2. Red Zone Identifier
3. Red Line Of Position
4. Red Master Line Status
5. Green Zone Identifier
6. Green Line Of Position
7. Green Master Line Status
8. Purple Zone Identifier
9. Purple Line Of Position
10. Purple Master Line Status
11. Red Line Navigation Use
12. Green Line Navigation Use
13. Purple Line Navigation Use
14. Position Uncertainty
15. N = Nautical Miles
16. Fix Data Basis
 - 1 = Normal Pattern
 - 2 = Lane Identification Pattern
 - 3 = Lane Identification Transmissions
17. Checksum

(The DCN sentence is obsolete as of 3.01)

DPT - Depth of Water

```
  1  2  3  4
  |  |  |  |
$--DPT,x.x,x.x,x.x*hh<CR><LF>
Field Number:
```

1. Water depth relative to transducer, meters
2. Offset from transducer, meters positive means distance from transducer to water line negative means distance from transducer to keel
3. Maximum range scale in use (NMEA 3.0 and above)
4. Checksum

This sentence was incorrectly titled "Heading - Deviation & Variation" in [BETKE]. It's documented at <http://www.humminbird.com/normal.asp?id=853>

Example: \$INDPT,2.3,0.0*46

DTM - Datum Reference

```
  1  2  3  4  5  6  7  8  9
  |  |  |  |  |  |  |  |
$ --DTM,ref,x,IIII,c,IIII,c,aaa,ref*hh<CR><LF>
Field Number:
```

1. Local datum code.
2. Local datum subcode. May be blank.
3. Latitude offset (minutes)
4. N or S
5. Longitude offset (minutes)
6. E or W
7. Altitude offset in meters
8. Datum name. What's usually seen here is "W84", the standard WGS84 datum used by GPS.
9. Checksum.

Example: \$GPDTM,W84,C*52

FSI - Frequency Set Information

Set (or report) frequency, mode of operation and transmitter power level of a radiotelephone.

1	2	3 4 5

\$--FSI,xxxxxx,xxxxxx,c,x*hh<CR><LF>

Field Number:

1. Transmitting Frequency
2. Receiving Frequency
3. Communications Mode (NMEA Syntax 2)
4. Power Level (0 = standby, 1 = lowest, 9 = highest)
5. Checksum

GBS - GPS Satellite Fault Detection

1	2	3	4	5	6	7	8	9

\$--GBS,hhmmss.ss,x.x,x.x,x.x,x.x,x.x,x.x,x.x*hh<CR><LF>

Field Number:

1. UTC time of the GGA or GNS fix associated with this sentence. hh is hours, mm is minutes, ss.ss is seconds
2. Expected 1-sigma error in latitude (meters)
3. Expected 1-sigma error in longitude (meters)
4. Expected 1-sigma error in altitude (meters)
5. ID of most likely failed satellite (1 to 138)
6. Probability of missed detection for most likely failed satellite
7. Estimate of bias in meters on most likely failed satellite
8. Standard deviation of bias estimate
9. Checksum

Note: Source [\[MX521\]](#) describes a proprietary extension of GBS with a 9th data field. The 8-field version is in NMEA 3.0.

Example: \$GPGBS,125027,23.43,M,13.91,M,34.01,M*07

GGA - Global Positioning System Fix Data

This is one of the sentences commonly emitted by GPS units.
Time, Position and fix related data for a GPS receiver.

															11
1	2	3 4	5 6 7 8	9 10		12 13	14	15							

\$--GGA,hhmmss.ss,ddmm.mm,a,ddmm.mm,a,x,xx,x.x,x.x,M,x.x,M,x.x,xxxx*hh<CR><LF>

Field Number:

1. UTC of this position report, hh is hours, mm is minutes, ss.ss is seconds.
2. Latitude, dd is degrees, mm.mm is minutes
3. N or S (North or South)
4. Longitude, dd is degrees, mm.mm is minutes
5. E or W (East or West)
6. GPS Quality Indicator (non null)
 - 0 - fix not available,
 - 1 - GPS fix,
 - 2 - Differential GPS fix (values above 2 are 2.3 features)
 - 3 = PPS fix
 - 4 = Real Time Kinematic
 - 5 = Float RTK
 - 6 = estimated (dead reckoning)
 - 7 = Manual input mode
 - 8 = Simulation mode
7. Number of satellites in use, 00 - 12
8. Horizontal Dilution of precision (meters)
9. Antenna Altitude above/below mean-sea-level (geoid) (in meters)
10. Units of antenna altitude, meters
11. Geoidal separation, the difference between the WGS-84 earth ellipsoid and mean-sea-level (geoid), "-" means mean-sea-level below ellipsoid
12. Units of geoidal separation, meters
13. Age of differential GPS data, time in seconds since last SC104 type 1 or 9 update, null field when DGPS is not used
14. Differential reference station ID, 0000-1023
15. Checksum

The number of digits past the decimal point for Time, Latitude and Longitude is model dependent.

Example:

\$GNGGA,001043.00,4404.14036,N,12118.85961,W,1,12,0.98,1113.0,M,-21.3,M*47

Note: Jackson Labs replaces the Quality indicator with GPSDO status.

GLC - Geographic Position, Loran-C

This sentence is obsolete over most of its former coverage area. The US/Canadian/Russian Loran-C network was shut down in 2010; it is still as of 2015 in

limited use in Europe. Loran-C operations in Norway will cease from 1st Jan 2016.

[NORWAY]

```

12 14
1 2 3 4 5 6 7 8 9 10 11| 13|
| | || || || || || ||
$--GLC,xxxx,x.x,a,x.x,a,x.x,a,x.x,a,x.x,a,x.x,a*hh<CR><LF>
Field Number:
```

1. GRI Microseconds/10
2. Master TOA Microseconds
3. Master TOA Signal Status
4. Time Difference 1 Microseconds
5. Time Difference 1 Signal Status
6. Time Difference 2 Microseconds
7. Time Difference 2 Signal Status
8. Time Difference 3 Microseconds
9. Time Difference 3 Signal Status
10. Time Difference 4 Microseconds
11. Time Difference 4 Signal Status
12. Time Difference 5 Microseconds
13. Time Difference 5 Signal Status
14. Checksum

GLL - Geographic Position - Latitude/Longitude

This is one of the sentences commonly emitted by GPS units.

```

1 2 3 4 5 6 7
| || || ||
$--GLL,ddmm.mm,a,dddmm.mm,a,hhmmss.ss,a*hh<CR><LF>
NMEA 2.3:
$--GLL,ddmm.mm,a,dddmm.mm,a,hhmmss.ss,a,m*hh<CR><LF>
Field Number:
```

1. Latitude, dd is degrees, mm.mm is minutes
2. N or S (North or South)
3. Longitude, dd is degrees, mm.mm is minutes
4. E or W (East or West)
5. UTC of this position, hh is hours, mm is minutes, ss.ss is seconds
6. Status A - Data Valid, V - Data Invalid
7. FAA mode indicator (NMEA 2.3 and later)
8. Checksum

The number of digits past the decimal point for Time, Latitude and Longitude is model dependent.

Example: \$GNGLL,4404.14012,N,12118.85993,W,001037.00,A,A*67

GNS - Fix data

1	2	3 4	5 6	7	8	9	10	11	12	13

\$--GNS,hhmmss.ss,ddmm.mm,a,dddmm.mm,a,c--c,xx,x.x,x.x,x.x,x.x,x.x*hh

Field Number:

1. UTC of position, hh is hours, mm is minutes, ss.ss is seconds.
2. Latitude, dd is minutes, mm.mm is minutes
3. N or S (North or South)
4. Longitude, dd is minutes, mm.mm is minutes
5. E or W (East or West)
6. Mode indicator (non-null)
7. Total number of satellites in use, 00-99
8. Horizontal Dilution of Precision, HDOP
9. Antenna altitude, meters, re:mean-sea-level(geoid).
10. Geoidal separation meters
11. Age of differential data
12. Differential reference station ID
13. Navigational status (optional) S = Safe C = Caution U = Unsafe V = Not valid for navigation
14. Checksum

The Mode indicator is one to four characters, with the first and second defined for GPS and GLONASS. Further characters may be defined. For each system, the character can have a value (table may be incomplete):

- A = Autonomous (non-differential)
- D = Differential mode
- E = Estimated (dead reckoning) Mode
- F = RTK Float
- M = Manual Input Mode
- N = Constellation not in use, or no valid fix
- P = Precise (no degradation, like Selective Availability, and hires)
- R = RTK Integer
- S = Simulator Mode

The number of digits past the decimal point for Time, Latitude and Longitude is model dependent.

Example: \$GPGNS,112257.00,3844.24011,N,00908.43828,W,AN,03,10.5,,*57

GRS - GPS Range Residuals

```

  1  2 3 4 5 6 7 8 9 10 11 12 13 14 15
  |  | | | | | | | | | | | | |
$ --GRS,hhmmss.ss,m,xx,xx,xx,xx,xx,xx,xx,xx,xx,xx*hh<CR><LF>
Field Number:
```

1. UTC time of associated GGA fix. hh is hours, mm is minutes, ss.ss is seconds.
2. 0 = Residuals used in GGA, 1 = residuals calculated after GGA
3. Satellite 1 residual in meters
4. Satellite 2 residual in meters
5. Satellite 3 residual in meters
6. Satellite 4 residual in meters (blank if unused)
7. Satellite 5 residual in meters (blank if unused)
8. Satellite 6 residual in meters (blank if unused)
9. Satellite 7 residual in meters (blank if unused)
10. Satellite 8 residual in meters (blank if unused)
11. Satellite 9 residual in meters (blank if unused)
12. Satellite 10 residual in meters (blank if unused)
13. Satellite 11 residual in meters (blank if unused)
14. Satellite 12 residual in meters (blank if unused)
15. System ID (NMEA 4.11), see above
16. Signal ID (NMEA 4.11), see above

xx. Checksum

The order of satellites MUST match those in the last GSA.

Example: \$GPGRS,024603.00,1,-1.8,-2.7,0.3,,,,,,*6C

Note that the talker ID may be GP, GL, or GN, depending on if the residuals are for GPS-only, GLONASS-only, or combined solution, respectively.

GST - GPS Pseudorange Noise Statistics

```

  1  2 3 4 5 6 7 8  9
  |  | | | | | | |
$ --GST,hhmmss.ss,x,x,x,x,x,x,x*hh<CR><LF>
Field Number:
```

1. TC time of associated GGA fix

2. Total RMS standard deviation of ranges inputs to the navigation solution
3. Standard deviation (meters) of semi-major axis of error ellipse
4. Standard deviation (meters) of semi-minor axis of error ellipse
5. Orientation of semi-major axis of error ellipse (true north degrees)
6. Standard deviation (meters) of latitude error
7. Standard deviation (meters) of longitude error
8. Standard deviation (meters) of altitude error
9. Checksum

Example: \$GPGST,182141.000,15.5,15.3,7.2,21.8,0.9,0.5,0.8*54

GSA - GPS DOP and active satellites

This is one of the sentences commonly emitted by GPS units.

```
1 2 3          14 15 16 17 18
| | |          | | | | |
```

\$--GSA,a,a,x,x,x,x,x,x,x,x,x,x,x,x,x,x,x,x,x,x,x*hh<CR><LF>

Field Number:

1. Selection mode: M=Manual, forced to operate in 2D or 3D, A=Automatic, 2D/3D
2. Mode (1 = no fix, 2 = 2D fix, 3 = 3D fix)
3. ID of 1st satellite used for fix
4. ID of 2nd satellite used for fix
5. ID of 3rd satellite used for fix
6. ID of 4th satellite used for fix
7. ID of 5th satellite used for fix
8. ID of 6th satellite used for fix
9. ID of 7th satellite used for fix
10. ID of 8th satellite used for fix
11. ID of 9th satellite used for fix
12. ID of 10th satellite used for fix
13. ID of 11th satellite used for fix
14. ID of 12th satellite used for fix
15. PDOP
16. HDOP
17. VDOP
18. System ID (NMEA 4.11), see above

xx. Checksum

Example: \$GNGSA,A,3,80,71,73,79,69,,,,,,,,,1.83,1.09,1.47*17

Note: NMEA 4.1+ systems (u-blox 9, Quectel LCD79) may emit an extra field, System ID, just before the checksum.

- 1 = GPS L1C/A, L2CL, L2CM
- 2 = GLONASS L1 OF, L2 OF
- 3 = Galileo E1C, E1B, E5 bl, E5 bQ
- 4 = BeiDou B1I D1, B1I D2, B2I D1, B2I D12

GSV - Satellites in view

This is one of the sentences commonly emitted by GPS units.

These sentences describe the sky position of a GPS satellite in view. Typically they're shipped in a group of 2 or 3.

```

1 2 3 4 5 6 7   n
| | | | | | |   |
$--GSV,x,x,x,x,x,x,x,...*hh<CR><LF>
Field Number:

```

1. total number of GSV sentences to be transmitted in this group
2. Sentence number, 1-9 of this GSV message within current group
3. total number of satellites in view (leading zeros sent)
4. satellite ID or PRN number (leading zeros sent)
5. elevation in degrees (-90 to 90) (leading zeros sent)
6. azimuth in degrees to true north (000 to 359) (leading zeros sent)
7. SNR in dB (00-99) (leading zeros sent) more satellite info quadruples like 4-7
n-1) Signal ID (NMEA 4.11) n) checksum

Example:

```

$GPGSV,3,1,11,03,03,111,00,04,15,270,00,06,01,010,00,13,06,292,00*74
$GPGSV,3,2,11,14,25,170,00,16,57,208,39,18,67,296,40,19,40,246,00*74
$GPGSV,3,3,11,22,42,067,42,24,14,311,43,27,05,244,00,,,*,4D

```

Some GPS receivers may emit more than 12 quadruples (more than three GPGSV sentences), even though NMEA-0813 doesn't allow this. (The extras might be WAAS satellites, for example.) Receivers may also report quads for satellites they aren't tracking, in which case the SNR field will be null; we don't know whether this is formally allowed or not.

Example: \$GLGSV,3,3,09,88,07,028*51

Note: NMEA 4.10+ systems (u-blox 9, Quectel LCD79) may emit an extra field, Signal ID, just before the checksum. See the description of Signal ID's above.

Note: \$GNGSV uses PRN in field 4. Other \$GxGSV use the satellite ID in field 4. Jackson Labs, Quectel, Telit, and others get this wrong, in various conflicting ways.

GTD - Geographic Location in Time Differences

```

1 2 3 4 5 6

```

| | | | |
 \$--GTD,x.x,x.x,x.x,x.x,x.x*hh<CR><LF>

Field Number:

1. time difference
2. time difference
3. time difference
4. time difference
5. time difference n) checksum

GXA - TRANSIT Position - Latitude/Longitude

Location and time of TRANSIT fix at waypoint

1 2 3 4 5 6 7 8
 | | || || ||
 \$--GXA,hhmmss.ss,ddmm.mm,a,dddmm.mm,a,c--c,X*hh<CR><LF>

Field Number:

1. UTC of position fix, hh is hours, mm is minutes, ss.ss is seconds.
2. Latitude, dd is degrees, mm.mm is minutes
3. East or West
4. Longitude, dd is degrees, mm.mm is minutes
5. North or South
6. Waypoint ID
7. Satellite number
8. Checksum

The number of digits past the decimal point for Time, Latitude and Longitude is model dependent.

(The GXA sentence is obsolete as of 3.01.)

HDG - Heading - Deviation & Variation

1 2 3 4 5 6
 | | || ||
 \$--HDG,x.x,x.x,a,x.x,a*hh<CR><LF>

Field Number:

1. Magnetic Sensor heading in degrees
2. Magnetic Deviation, degrees
3. Magnetic Deviation direction, E = Easterly, W = Westerly
4. Magnetic Variation degrees
5. Magnetic Variation direction, E = Easterly, W = Westerly

6. Checksum

HDM - Heading - Magnetic

Vessel heading in degrees with respect to magnetic north produced by any device or system producing magnetic heading.

1 2 3

| ||

\$--HDM,x.x,M*hh<CR><LF>

Field Number:

1. Heading Degrees, magnetic
2. M = magnetic
3. Checksum

HDT - Heading - True

Actual vessel heading in degrees true produced by any device or system producing true heading.

Deprecated 2008.

1 2 3

| ||

\$--HDT,x.x,T*hh<CR><LF>

Field Number:

1. Heading, degrees True
2. T = True
3. Checksum

Example: \$GPHDT,274.07,T*03

HFB - Trawl Headrope to Footrope and Bottom

1 2 3 4 5

| | | |

\$--HFB,x.x,M,y.y,M*hh<CR><LF>

Field Number:

1. Distance from headrope to footrope
2. Meters (0-100)
3. Distance from headrope to bottom
4. Meters (0-100)
5. Checksum

From [GLOBALSAT]. Shown with a "@ll" leader rather than "\$GP".

HSC - Heading Steering Command

```

      1 2 3 4 5
      | | | |
$--HSC,x,x,T,x,x,M*hh<CR><LF>
      Field Number:

```

1. Heading Degrees, True
2. T = True
3. Heading Degrees, Magnetic
4. M = Magnetic
5. Checksum

[GLOBALSAT] describes a completely different meaning for this sentence, having to do with water temperature sensors. It is unclear which is correct.

HWBIAS - Unknown

Unknown sentence. Seen from Huawei GPS receivers.

Example: \$GPHWBIAS,11.2*21

ITS - Trawl Door Spread 2 Distance

1 2 3
| | |
\$--ITS,x.x,M*hh<CR><LF>
Field Number:

1. Second spread distance
2. Meters
3. Checksum.

From [GLOBSAT]. Shown with a "@ll" leader rather than "\$GP".

LCD - Loran-C Signal Data

Obsolete.

1	2	3	4	5	6	7	8	9	10	11	12	13	14
\$--LCD,xxxx,xxx,xxx,xxx,xxx,xxx,xxx,xxx,xxx,xxx,xxx,xxx,xxx*hh<CR><LF>													
Field Number:													

1. GRI Microseconds/10
2. Master Relative SNR
3. Master Relative ECD
4. Time Difference 1 Microseconds
5. Time Difference 1 Signal Status
6. Time Difference 2 Microseconds
7. Time Difference 2 Signal Status
8. Time Difference 3 Microseconds
9. Time Difference 3 Signal Status
10. Time Difference 4 Microseconds
11. Time Difference 4 Signal Status
12. Time Difference 5 Microseconds
13. Time Difference 5 Signal Status
14. Checksum

MDA - Meteorological Composite

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21
 | | | | | | | | | | | | | | | | |
 \$--MDA,n.nn,I,n.nnn,B,n.n,C,n.C,n.n,n,n.n,C,n.n,T,n.n,M,n.n,N,n.n,M*hh<CR><LF>
 Field Number:

1. Barometric pressure, inches of mercury, to the nearest 0.01 inch
2. I = inches of mercury
3. Barometric pressure, bars, to the nearest .001 bar
4. B = bars
5. Air temperature, degrees C, to the nearest 0.1 degree C
6. C = degrees C
7. Water temperature, degrees C (this field left blank by WeatherStation)
8. C = degrees C
9. Relative humidity, percent, to the nearest 0.1 percent
10. Absolute humidity, percent
11. Dew point, degrees C, to the nearest 0.1 degree C
12. C = degrees C
13. Wind direction, degrees True, to the nearest 0.1 degree
14. T = true
15. Wind direction, degrees Magnetic, to the nearest 0.1 degree
16. M = magnetic
17. Wind speed, knots, to the nearest 0.1 knot
18. N = knots
19. Wind speed, meters per second, to the nearest 0.1 m/s

20. M = meters per second

21. Checksum

Obsolete as of 2009.

MSK - Control for a Beacon Receiver

1 2 3 4 5 6

| | | | | |

\$--MSK,nnn,m,nnn,m,nnn*hh<CR><LF>

Field Number:

1. Beacon frequency to use, 283.5-325.0 kHz
2. Frequency mode, A=auto, M=manual
3. Beacon bit rate (25, 50, 100, 200)
4. Bitrate, A=auto, M=manual
5. Interval for MSS message status (null for no status), seconds
6. Checksum

MSS - Beacon Receiver Status

1 2 3 4 5 6

| | | | | |

\$--MSS,nn,nn,fff,bbb,xxx*hh<CR><LF>

Field Number:

1. Signal strength (dB 1uV)
2. Signal to noise ratio (dB)
3. Beacon frequency, 283.5-325.0 kHz
4. Beacon data rate, 25, 50, 100, 200 bps
5. Channel number
6. Checksum

Example: \$GPMSS,0,0,0.000000,200*5A

MTW - Mean Temperature of Water

1 2 3

| | |

\$--MTW,x.x,C*hh<CR><LF>

Field Number:

1. Temperature, degrees
2. Unit of Measurement, Celsius

3. Checksum

[[GLOBALSAT](#)] lists this as "Meteorological Temperature of Water", which is probably incorrect.

Example: \$INMTW,17.9,C*1B

MWD - Wind Direction & Speed

1 2 3 4 5 6 7 8 9

| || || || ||

\$--MWD,x.x,a,x.x,a,x.x,a,x.x,a*hh<CR><LF>

Field Number:

1. Wind Direction, 0 to 359.9 degrees
2. Reference, T = True, M = Magnetic
3. Wind Direction, 0 to 359.9 degrees
4. Reference, T = True, M = Magnetic
5. Wind Speed
6. Wind Speed Units, K/M/N
7. Wind Speed
8. Wind Speed Units, K/M/N
9. Checksum

Example: \$WIMWD,302.4,T,289.6,M,10.5,N,5.4,M*6F

MWV - Wind Speed and Angle

1 2 3 4 5

| || ||

\$--MWV,x.x,a,x.x,a*hh<CR><LF>

Field Number:

1. Wind Angle, 0 to 359 degrees
2. Reference, R = Relative, T = True
3. Wind Speed
4. Wind Speed Units, K/M/
5. Status, A = Data Valid, V = Invalid
6. Checksum

OLN - Omega Lane Numbers

Obsolete.

1 2 3 4

```

|-----+ |-----+ |-----+ |
$--OLN,aa,xxx,xxx,aa,xxx,xxx,aa,xxx,xxx*hh<CR><LF>
Field Number:

```

1. Omega Pair 1
2. Omega Pair 1
3. Omega Pair 1
4. Checksum

(The OLN sentence is obsolete as of 2.30)

OSD - Own Ship Data

```

1 2 3 4 5 6 7 8 9 10
| || || || | ||
$--OSD,x.x,A,x.x,a,x.x,a,x.x,x.x,a*hh<CR><LF>
Field Number:

```

1. Heading, degrees True
2. Status, A = Data Valid, V = Invalid
3. Vessel Course, degrees True
4. Course Reference B/M/W/R/P
5. Vessel Speed
6. Speed Reference B/M/W/R/P
7. Vessel Set, degrees True
8. Vessel drift (speed)
9. Speed Units K/N
10. Checksum

R00 - Waypoints in active route

```

1          n
|          |
$--R00,c---c,c---c,...*hh<CR><LF>
Field Number:

```

1. waypoint ID
- ...
- n) checksum

RLM – Return Link Message

```

1          2      3 4 5

```

```

      |      |      | | |
$--RLM,hhhhhhhhhhhhhhhh,hhmmss.ss,h,h--h*hh<CR><LF>

```

Field Number:

1. Beacon ID
2. Time of Reception, hh is hours, mm is minutes, ss.ss is seconds.
3. Message Code
4. Message Body
5. Checksum

RMA - Recommended Minimum Navigation Information

```

                        12
      1 2   3 4   5 6 7 8 9 10 11|
      ||   ||   || | | | | |
$--RMA,A,III.II,a,yyyyy.yy,a,x.x,x.x,x.x,x.x,a*hh<CR><LF>

```

Field Number:

1. Status, A = Valid, V = Invalid
2. Latitude
3. N or S
4. Longitude
5. E or W
6. Time Difference A, uS
7. Time Difference B, uS
8. Speed Over Ground, Knots
9. Track Made Good, degrees True
10. Magnetic Variation, degrees
11. E or W
12. Checksum

RMB - Recommended Minimum Navigation Information

To be sent by a navigation receiver when a destination waypoint is active.

```

                        14
      1 2 3 4 5 6   7 8   9 10 11 12 13|
      || || | |   ||   || | | |
$--RMB,A,x.x,a,c--c,c--c,III.II,a,yyyyy.yy,a,x.x,x.x,x.x,A*hh<CR><LF>

```

NMEA 2.3:

```

$--RMB,A,x.x,a,c--c,c--c,III.II,a,yyyyy.yy,a,x.x,x.x,x.x,A,m*hh<CR><LF>

```

Field Number:

1. Status, A = Active, V = Invalid
2. Cross Track error - nautical miles
3. Direction to Steer, Left or Right
4. Origin Waypoint ID
5. Destination Waypoint ID
6. Destination Waypoint Latitude
7. N or S
8. Destination Waypoint Longitude
9. E or W
10. Range to destination in nautical miles
11. Bearing to destination in degrees True
12. Destination closing velocity in knots
13. Arrival Status, A = Arrival Circle Entered. V = not entered/passed
14. FAA mode indicator (NMEA 2.3 and later)
15. Checksum

Example:

\$GPRMB,A,0.66,L,003,004,4917.24,N,12309.57,W,001.3,052.5,000.5,V*0B

RMC - Recommended Minimum Navigation Information

This is one of the sentences commonly emitted by GPS units.

1	2 3	4 5	6 7	8 9	10 11

\$--RMC,hhmmss.ss,A,ddmm.mm,a,dddmm.mm,a,x.x,x.x,xxxx,x.x,a*hh<CR><LF>

NMEA 2.3:

\$--RMC,hhmmss.ss,A,ddmm.mm,a,dddmm.mm,a,x.x,x.x,xxxx,x.x,a,m*hh<CR><LF>

NMEA 4.1:

\$--RMC,hhmmss.ss,A,ddmm.mm,a,dddmm.mm,a,x.x,x.x,xxxx,x.x,a,m,s*hh<CR><LF>

Field Number:

1. UTC of position fix, hh is hours, mm is minutes, ss.ss is seconds.
2. Status, A = Valid, V = Warning
3. Latitude, dd is degrees. mm.mm is minutes.
4. N or S
5. Longitude, ddd is degrees. mm.mm is minutes.
6. E or W
7. Speed over ground, knots
8. Track made good, degrees true
9. Date, ddmmyy
10. Magnetic Variation, degrees
11. E or W

12. FAA mode indicator (NMEA 2.3 and later)
13. Nav Status (NMEA 4.1 and later) A=autonomous, D=differential, E=Estimated, M=Manual input mode N=not valid, S=Simulator, V = Valid
14. Checksum

A status of V means the GPS has a valid fix that is below an internal quality threshold, e.g. because the dilution of precision is too high or an elevation mask test failed.

The number of digits past the decimal point for Time, Latitude and Longitude is model dependent.

Example:

\$GNRMC,001031.00,A,4404.13993,N,12118.86023,W,0.146,,100117,,,A*7B

ROT - Rate Of Turn

1 2 3

| | |

\$--ROT,x.x,A*hh<CR><LF>

Field Number:

1. Rate Of Turn, degrees per minute, "-" means bow turns to port
2. Status, A means data is valid
3. Checksum

Example: \$HEROT,0.0,A*2B

RPM - Revolutions

1 2 3 4 5 6

| | | | | |

\$--RPM,a,x,x.x,x.x,A*hh<CR><LF>

Field Number:

1. Source, S = Shaft, E = Engine
2. Engine or shaft number
3. Speed, Revolutions per minute
4. Propeller pitch, % of maximum, "-" means astern
5. Status, A = Valid, V = Invalid
6. Checksum

RSA - Rudder Sensor Angle

1 2 3 4 5

```

| | | |
$--RSA,x.x,A,x.x,A*hh<CR><LF>

```

Field Number:

1. Starboard (or single) rudder sensor, "-" means Turn To Port
2. Status, A = valid, V = Invalid
3. Port rudder sensor
4. Status, A = valid, V = Invalid
5. Checksum

RSD - RADAR System Data

```

14
1 2 3 4 5 6 7 8 9 10 11 12 13|
| | | | | | | | | | | |
$--RSD,x.x,x.x,x.x,x.x,x.x,x.x,x.x,x.x,x.x,x.x,x.x,x.x,a,a*hh<CR><LF>

```

(Some fields are missing from this description.)

Field Number:

1. Origin 1 range
2. Origin 1 bearing
3. Variable Range Marker 1
4. Bearing Line 1
5. Origin 2 range
6. Origin 2 bearing
7. Variable Range Marker 2
8. Bearing Line 2
9. Cursor Range From Own Ship
10. Cursor Bearing Degrees Clockwise From Zero
11. Range Scale
12. Range Units
13. Display rotation (C = course up, H = heading up, N - North up)
14. Checksum

RTE - Routes

```

1 2 3 4 5      x  n
| | | | |      |  |
$--RTE,x.x,x.x,a,c--c,c--c, ..... c--c*hh<CR><LF>

```

Field Number:

1. Total number of RE sentences being transmitted

2. Sentence Number
3. Sentence mode c = complete route, all waypoints w = working route, the waypoint you just left, the waypoint you're heading to, then all the rest
4. Route ID
5. Waypoint ID
6. Additional waypoint IDs

More waypoints follow. Last field is a checksum as usual.

The Garmin 65 and possibly other units report a \$GPR00 in the same format.

Example: \$GPRTE,1,1,c,0*07

SFI - Scanning Frequency Information

1	2	3	4		x

\$--SFI,x.x,x.x,x,xxxxxx,c xxxxxx,c*hh<CR><LF>

Field Number:

1. Total Number Of sentences
2. Sentence Number
3. Frequency 1
4. Mode 1
5. Additional Frequency, Mode pairs
6. Checksum

STN - Multiple Data ID

This sentence is transmitted before each individual sentence where there is a need for the Listener to determine the exact source of data in the system. Examples might include dual-frequency depth sounding equipment or equipment that integrates data from a number of sources and produces a single output.

1	2

\$--STN,x.x*hh<CR><LF>

Field Number:

1. Talker ID Number
2. Checksum

TDS - Trawl Door Spread Distance

1	2	3

\$--TDS,x.x,M*hh<CR><LF>

Field Number:

1. Distance between trawl doors
2. Meters (0-300)
3. Checksum.

From [\[GLOBALSAT\]](#). Shown with a "@II" leader rather than "\$GP".

TFI - Trawl Filling Indicator

1 2 3 4

| | | |

\$--TFI,x,y,z*hh<CR><LF>

Field Number:

1. Catch sensor #1 (0 = off, 1 = on, 2 = no answer)
2. Catch sensor #2 (0 = off, 1 = on, 2 = no answer)
3. Catch sensor #3 (0 = off, 1 = on, 2 = no answer)

From [\[GLOBALSAT\]](#). Shown with a "@II" leader rather than "\$GP".

TLB - Target Label

1 2 (3) (4) 5

| | | | |

\$--TLB,x.x,c--c,x.x,c--c,...,x.x,c--c*hh<CR><LF>

Field Number:

1. Target Number (0-99)
2. Label assigned to target
3. (Target number 2)
4. (Label assigned to target number 2)
5. Checksum

Message can contain target number + label pairs up to maximum NMEA message length. Target number references to target number in TTM (and/or TLL) messages.

TLL - Target Latitude and Longitude

1 2 3 4 5 6 7 8 9 10

| | | | | | | |

\$--TLL,xx,ddmm.mm,a,dddmm.mm,a,c--c,hhmmss.ss,a,a*hh<CR><LF>

Field Number:

1. Target Number (0-99)
2. Target Latitude, dd is degrees, mm.mm is minutes.
3. N=north, S=south
4. Target Longitude, dd is degrees, mm.mm is minutes.
5. E=east, W=west
6. Target name
7. UTC of data, hh is hours, mm is minutes, ss.ss is seconds.
8. Status (L=lost, Q=acquisition, T=tracking)
9. R= reference target; null (,,)= otherwise

The number of digits past the decimal point for Time, Latitude and Longitude is model dependent.

TPC - Trawl Position Cartesian Coordinates

```

1 2 3 4 5 6 7
| | | | | | |
$--TPC,x,M,y,P,z.z,M*hh,<CR><LF>
Field Number:

```

1. Horizontal distance from the vessel center line
2. Meters
3. Horizontal distance from the transducer to the trawl along the vessel center line.
The value is normally positive assuming the trawl is located behind the vessel.
4. Meters
5. Depth of the trawl below the surface
6. Meters
7. Checksum

From [\[GLOBALSAT\]](#). Shown with a "@II" leader rather than "\$GP". This entry actually merges their TPC description with another entry labeled (apparently incorrectly) TPT, which differs from the TPT shown below.

TPR - Trawl Position Relative Vessel

```

1 2 3 4 5 6 7
| | | | | | |
$--TPR,x,M,y,P,z.z,M*hh,<CR><LF>
Field Number:

```

1. Horizontal range relative to target
2. Meters (0-4000)
3. Bearing to target relative to vessel heading. Resolution is one degree.

4. Separator
5. Depth of trawl below the surface
6. Meters (0-2000)
7. Checksum

From [\[GLOBALSAT\]](#). Shown with a "@II" leader rather than "\$GP".

TPT - Trawl Position True

1 2 3 4 5 6 7

| | | | | | |

\$--TPT,x,M,y,P,z.z,M*hh,<CR><LF>

Field Number:

1. Horizontal range relative to target
2. Meters (0-4000)
3. True bearing to target (ie. relative north). Resolution is one degree.
4. Separator
5. Depth of trawl below the surface
6. Meters (0-2000)
7. Checksum

From [\[GLOBALSAT\]](#). Shown with a "@II" leader rather than "\$GP".

TRF - TRANSIT Fix Data

													13
1	2	3	4	5	6	7	8	9	10	11	12		

\$--TRF,hhmmss.ss,xxxxxx,ddmm.mm,a,dddmm.mm,a,x.x,x.x,x.x,x.x,xxx,A*hh<CR><LF>

Field Number:

1. UTC of fix, hh is hours, mm is minutes, ss.ss is seconds.
2. Date, ddmmyy
3. Latitude, dd is degrees, mm.mm is minutes.
4. N or S
5. Longitude, dd is degrees, mm.mm is minutes.
6. E or W
7. Elevation Angle
8. Number of iterations
9. Number of Doppler intervals

10. Update distance, nautical miles
11. Satellite ID
12. Data Validity
13. Checksum

The number of digits past the decimal point for Time, Latitude and Longitude is model dependent.

(The TRF sentence is obsolete as of 2.3.0)

TTM - Tracked Target Message

11											13		16			
1	2	3	4	5	6	7	8	9	10	12	14	15				

\$--TTM,xx,x.x,x.x,a,x.x,x.x,a,x.x,x.x,a,c--c,a,a,hhmmss.ss,a*hh<CR><LF>

Field Number:

1. Target Number (0-99)
2. Target Distance
3. Bearing from own ship
4. T = True, R = Relative
5. Target Speed
6. Target Course
7. T = True, R = Relative
8. Distance of closest-point-of-approach
9. Time until closest-point-of-approach "-" means increasing
10. Speed/distance units, K/N
11. Target name
12. Target Status
13. Reference Target
14. UTC of data (NMEA 3 and above) hh is hours, mm is minutes, ss.ss is seconds.
15. Type, A = Auto, M = Manual, R = Reported (NMEA 3 and above)
16. Checksum

[GLOBALSAT] gives this in a slightly different form, with 14th and 15th fields conveying time of observation and whether target acquisition was automatic or manual.

VBW - Dual Ground/Water Speed

```

      1 2 3 4 5 6 7 8 0 10 11
      | | || | || || |
$--VBW,x,x,x,x,A,x,x,x,x,A,x,x,A,x,x,A*hh<CR><LF>

```

Field Number:

1. Longitudinal water speed, "-" means astern, knots
2. Transverse water speed, "-" means port, knots
3. Status, A = Data Valid
4. Longitudinal ground speed, "-" means astern, knots
5. Transverse ground speed, "-" means port, knots
6. Status, A = Data Valid
7. Stern traverse water speed, knots *NMEA 3 and above)
8. Status, stern traverse water speed A = Valid (NMEA 3 and above)
9. Stern traverse ground speed, knots *NMEA 3 and above)
10. Status, stern ground speed A = Valid (NMEA 3 and above)
11. Checksum

VDR - Set and Drift

1 2 3 4 5 6 7
| | | | |

\$--VDR,x.x,T,x.x,M,x.x,N*hh<CR><LF>

Field Number:

1. Direction degrees, True
2. T = True
3. Direction degrees Magnetic
4. M = Magnetic
5. Current speed, knots
6. N = Knots
7. Checksum

VHW - Water speed and heading

1 2 3 4 5 6 7 8 9
| | | | | |

\$--VHW,x.x,T,x.x,M,x.x,N,x.x,K*hh<CR><LF>

Field Number:

1. Heading degrees, True
2. T = True
3. Heading degrees, Magnetic
4. M = Magnetic
5. Speed of vessel relative to the water, knots
6. N = Knots

7. Speed of vessel relative to the water, km/h
8. K = Kilometers
9. Checksum

[GLOBALSAT] describes a different format in which the first three fields are water-temperature measurements. It's not clear which is correct.

VLW - Distance Traveled through Water

```

1 2 3 4 5 6 7 8 9
| | | | |
$--VLW,x.x,N,x.x,N,x.x,N,x.x,N*hh<CR><LF>
Field Number:

```

1. Total cumulative water distance, nm
2. N = Nautical Miles
3. Water distance since Reset, nm
4. N = Nautical Miles
5. Total cumulative ground distance, nm (NMEA 3 and above)
6. N = Nautical Miles (NMEA 3 and above)
7. Ground distance since reset, nm (NMEA 3 and above)
8. N = Nautical Miles (NMEA 3 and above)
9. Checksum

VPW - Speed - Measured Parallel to Wind

```

1 2 3 4 5
| | |
$--VPW,x.x,N,x.x,M*hh<CR><LF>
Field Number:

```

1. Speed, "-" means downwind, knots
2. N = Knots
3. Speed, "-" means downwind, m/s
4. M = Meters per second
5. Checksum

VTG - Track made good and Ground speed

This is one of the sentences commonly emitted by GPS units.

```

1 2 3 4 5 6 7 8 9
| | | | |
$--VTG,x.x,T,x.x,M,x.x,N,x.x,K*hh<CR><LF>

```

NMEA 2.3:

\$--VTG,x.x,T,x.x,M,x.x,N,x.x,K,m*hh<CR><LF>

Field Number:

1. Course over ground, degrees True
2. T = True
3. Course over ground, degrees Magnetic
4. M = Magnetic
5. Speed over ground, knots
6. N = Knots
7. Speed over ground, km/h
8. K = Kilometers Per Hour
9. FAA mode indicator (NMEA 2.3 and later)
10. Checksum

Note: in some older versions of NMEA 0183, the sentence looks like this:

1 2 3 4 5

| | | | |

\$--VTG,x.x,x,x,x,x,x*hh<CR><LF>

Field Number:

1. True course over ground (degrees) 000 to 359
2. Magnetic course over ground 000 to 359
3. Speed over ground (knots) 00.0 to 99.9
4. Speed over ground (kilometers) 00.0 to 99.9
5. Checksum

The two forms can be distinguished by field 2, which will be the fixed text 'T' in the newer form. The new form appears to have been introduced with NMEA 3.01 in 2002.

Some devices, such as those described in [\[GLOBALSAT\]](#), leave the magnetic-bearing fields 3 and 4 empty.

Example: \$GPVTG,220.86,T,,M,2.550,N,4.724,K,A*34

VWR - Relative Wind Speed and Angle

1 2 3 4 5 6 7 8 9

| | | | | | | | |

\$--VWR,x.x,a,x.x,N,x.x,M,x.x,K*hh<CR><LF>

Field Number:

1. Wind direction magnitude in degrees
2. Wind direction Left/Right of bow

3. Speed
4. N = Knots
5. Speed
6. M = Meters Per Second
7. Speed
8. K = Kilometers Per Hour
9. Checksum

WCV - Waypoint Closure Velocity

1 2 3 4 5

| || ||

\$--WCV,x.x,N,c--c,a*hh<CR><LF>

Field Number:

1. Velocity, knots
2. N = knots
3. Waypoint ID
4. FAA Mode indicator, not null (NMEA 3 and above)
5. Checksum

WNC - Distance - Waypoint to Waypoint

1 2 3 4 5 6 7

| || || | |

\$--WNC,x.x,N,x.x,K,c--c,c--c*hh<CR><LF>

Field Number:

1. Distance, Nautical Miles
2. N = Nautical Miles
3. Distance, Kilometers
4. K = Kilometers
5. TO Waypoint
6. FROM Waypoint
7. Checksum

WPL - Waypoint Location

1 2 3 4 5 6

| || || |

\$--WPL,III.II,a,yyyyy.yy,a,c--c*hh<CR><LF>

Field Number:

1. Latitude
2. N or S (North or South)
3. Longitude
4. E or W (East or West)
5. Waypoint name
6. Checksum

XDR - Transducer Measurement

```

1 2 3 4      n
|| ||      |
$--XDR,a,x.x,a,c--c, ..... *hh<CR><LF>
Field Number:

```

1. Transducer Type
2. Measurement Data
3. Units of measurement A = Amperes B = Bars B = Binary C = Celsius D = Degrees H = Hertz I = liters/second K = Kelvin K = kg/m³ M = Meters M = cubic Meters N = Newton P = % of full range P = Pascal R = RPM S = Parts per thousand V = Volts
4. Name of transducer

There may be any number of quadruplets like this, each describing a sensor. The last field will be a checksum as usual.

Example:

```

$HCXDR,A,171,D,PITCH,A,-37,D,ROLL,G,367,,MAGX,G,2420,,MAGY,G,-8984,,M
AGZ*41

```

XTE - Cross-Track Error, Measured

```

1 2 3 4 5 6
||| |||
$--XTE,A,A,x.x,a,N*hh<CR><LF>
NMEA 2.3:
$--XTE,A,A,x.x,a,N,m*hh<CR><LF>
Field Number:

```

1. Status
 - A - Valid
 - V = Loran-C Blink or SNR warning
 - V = general warning flag or other navigation systems when a reliable fix is not available

2. Status
 - V = Loran-C Cycle Lock warning flag
 - A = Valid
3. Cross Track Error Magnitude
4. Direction to steer, L or R
5. Cross Track Units, N = Nautical Miles
6. FAA mode indicator (NMEA 2.3 and later, optional)
7. Checksum

Example: \$GPXTE,V,V,,,N,S*43

XTR - Cross Track Error - Dead Reckoning

```

1  2 3 4
|  | | |
$--XTR,x.x,a,N*hh<CR><LF>
Field Number:

```

1. Magnitude of cross track error
2. Direction to steer, L or R
3. Units, N = Nautical Miles
4. Checksum

ZDA - Time & Date - UTC, day, month, year and local time zone

This is one of the sentences commonly emitted by GPS units.

```

1      2 3 4  5 6 7
|      | | |  | | |
$--ZDA,hhmmss.ss,xx,xx,xxxx,xx,xx*hh<CR><LF>
Field Number:

```

1. UTC time (hours, minutes, seconds, may have fractional subseconds)
2. Day, 01 to 31
3. Month, 01 to 12
4. Year (4 digits)
5. Local zone description, 00 to +- 13 hours
6. Local zone minutes description, 00 to 59, apply same sign as local hours
7. Checksum

Example: \$GPZDA,160012.71,11,03,2004,-1,00*7D

ZFO - UTC & Time from origin Waypoint

1	2	3	4

\$--ZFO,hhmmss.ss,hhmmss.ss,c--c*hh<CR><LF>
Field Number:

1. Universal Time Coordinated (UTC) hh is hours, mm is minutes, ss.ss is seconds.
2. Elapsed Time
3. Origin Waypoint ID
4. Checksum

ZTG - UTC & Time to Destination Waypoint

1	2	3	4

\$--ZTG,hhmmss.ss,hhmmss.ss,c--c*hh<CR><LF>
Field Number:

1. UTC of observation hh is hours, mm is minutes, ss.ss is seconds.
2. Time Remaining
3. Destination Waypoint ID
4. Checksum

Other sentences

There is evidence for the existence of the following NMEA sentences on the Web:

ACK - Alarm Acknowledgement

ADS - Automatic Device Status

AKD - Acknowledge Detail Alarm Condition

ALA - Set Detail Alarm Condition

ASD - Autopilot System Data

BEC - Bearing & Distance to Waypoint - Dead Reckoning

CEK - Configure Encryption Key Command

COP - Configure the Operational Period, Command

CUR - Water Current Layer

DCR - Device Capability Report

DDC - Display Dimming Control

DOR - Door Status Detection

DSC - Digital Selective Calling Information

DSE - Extended DSC

DSI - DSC Transponder Initiate

DSR - DSC Transponder Response

ETL - Engine Telegraph Operation Status

EVE - General Event Message

FIR - Fire Detection

WDR - Distance to Waypoint - Rhumb Line

WDC - Distance to Waypoint - Great Circle

ZDL - Time and Distance to Variable Point

\$CDDSC is described in [\[CDDSC\]](#).

Vendor extensions

This list is very incomplete.

PASHR - RT300 proprietary roll and pitch sentence

1	2	3	4	5	6	7	8	9	10	11	12

\$PASHR,hhmmss.sss,hhh.hh,T,rrr.rr,ppp.pp,xxx.xx,a.aaa,b.bbb,c.ccc,d,e*hh<CR><LF>

Field Number:

1. UTC time. mm is minutes, ss.sss is seconds.
2. hhh.hh - Heading in degrees
3. T - flag to indicate that the Heading is True Heading (i.e. to True North)
4. rrr.rr - Roll Angle in degrees
5. ppp.pp - Pitch Angle in degrees
6. xxx.xx - Heave
7. a.aaa - Roll Angle Accuracy Estimate (Stdev) in degrees

8. b.bbb - Pitch Angle Accuracy Estimate (Stdev) in degrees
9. c.ccc - Heading Angle Accuracy Estimate (Stdev) in degrees
10. d - Aiding Status
11. e - IMU Status
12. hh - Checksum

[PASHR] describes this sentence as NMEA, though other sources say it is Ashtech proprietary and describe a different format.

Example:

\$PASHR,085335.000,224.19,T,-01.26,+00.83,+00.00,0.101,0.113,0.267,1,0*06

PGLOR - Quectel

Undocumented Quectel messages. Found in LC79D GNSS.

- \$PGLOR,,AGC - ??
- \$PGLOR,,CPU - CPU Loading
- \$PGLOR,,FIN - Request completion status
- \$PGLOR,,FIX - Time To Fix
- \$PGLOR,,FTS - Factory Test Status
- \$PGLOR,,GFC - GeoFence Fix
- \$PGLOR,,GLO - ??
- \$PGLOR,,HLA - Value of HULA sensors
- \$PGLOR,,IMS - IMES messages
- \$PGLOR,,LSQ - Least squares GNSS fix
- \$PGLOR,NET - Report network information
- \$PGLOR,,NEW - Indicate new GPS request
- \$PGLOR,,PFM - Platform Status
- \$PGLOR,,PPS - Indicate PPS time corrections
- \$PGLOR,,PWR - Power consumption report
- \$PGLOR,,RID - Version Information
- \$PGLOR,,SAT - GPS Satellite information
- \$PGLOR,,SIO - Serial I/O status report
- \$PGLOR,,SPA - Spectrum analyzer results
- \$PGLOR,,SPD - ??
- \$PGLOR,SPL - ??
- \$PGLOR,,SPS - ??
- \$PGLOR,,STA - GLL status
- \$PGLOR,,SVC - ??
- \$PGLOR,,SVD - SV Dopplers detected in the false alarm test.
- \$PGLOR,,SMx - Report GPS Summary Information
- \$PGLOR,,UNC - ??

PGRME - Garmin Estimated Error

1 2 3 4 5 6 7

| | | | | | |

\$PGRME,hhh,M,vvv,M,ttt,M*hh<CR><LF>

Field Number:

1. Estimated horizontal position error (HPE),
2. M=meters
3. Estimated vertical position error (VPE)
4. M=meters
5. Overall spherical equivalent position error
6. M=meters
7. Checksum

Example: \$PGRME,15.0,M,45.0,M,25.0,M*22

PGRMZ - Garmin Altitude

1 2 3 4

| | | |

\$PGRMZ,hhh,f,M*hh<CR><LF>

Field Number:

1. Current Altitude Feet
2. f=feet
3. Mode (1 = no fix, 2 = 2D fix, 3 = 3D fix)
4. Checksum

Example: \$PGRMZ,2282,f,3*21

PJLTS - Jackson Labs GPSDO Status

1 2 3 4 5 6 7 8 9 10

| | | | | | | | | |

\$PJLTS,aaa.aa,bbb.bb,ccc,d,e.fffffff,ff.fff,g.gEhh,iii,jj,kkkk*hh<CR><LF>

Field Number:

1. Filtered UTC offset in ns
2. Raw UTC offset in ns from the time interval counter,
3. Number of captured input 1PPS signals,
4. Lock status
5. EFC voltage

6. EFC percentage
7. Estimated frequency accuracy
8. Seconds in holdover
9. Number of satellites tracked.
10. Health status

PJLTS - Jackson Labs Time and 3D velocity

1	2	3	4	5	6	7

\$PJLTV,aaaaa,bbbbbb,ccccc,ddd,eeeeeeee,ffff,gg*hh<CR><LF>

Field Number:

1. X velocity (cm/s)
2. Y velocity (cm/s)
3. Z velocity (cm/s)
4. speed accuracy estimate (cm/s)
5. GPS TOW in seconds
6. GPS week number
7. UTC leap second offset in seconds

PMGNST - Magellan Status

1	2	3	4	5	6	7	8

\$PMGNST,xx.xx,m,t,nnn,xx.xx,nnn,nn,c

Field Number:

1. Firmware version number?
2. Mode (1 = no fix, 2 = 2D fix, 3 = 3D fix)
3. T if we have a fix
4. Numbers change - unknown
5. Time left on the GPS battery in hours
6. Numbers change (freq. compensation?)
7. PRN number receiving current focus
8. Checksum

Only supported on Magellan receivers.

PRWIZCH - Rockwell Channel Status

\$PRWIZCH,n,s,n,s,n,s,n,s,n,s,n,s,n,s,n,s,n,s,n,s,c*hh<CR><LF>

Fields consist of 12 pairs of a satellite PRN followed by a signal quality number in the range 0-7 (0 worst, 7 best).

Only emitted by the now-obsolete Zodiac (Rockwell) chipset.

PUBX 00 - u-blox Lat/Long Position Data

\$PUBX,00,hhmmss.ss,Latitude,N,Longitude,E,AltRef,NavStat,Hacc,Vacc,SOG,COG,Vvel,+ageC,HDOP,VDOP,TDOP,GU,RU,DR*hh<CR><LF>

Example:

\$PUBX,00,081350.00,4717.113210,N,00833.915187,E,546.589,G3,2.1,2.0,0.007,77.52,0+.007,,0.92,1.19,0.77,9,0,0*5F<CR><LF>

Only emitted by u-blox Antaris chipset.

PUBX 01 - u-blox UTM Position Data

The \$PUBX,01 is a UTM (Universal Transverse Mercator projection) version of the \$PUBX,00 sentence.

\$PUBX,01,hhmmss.ss,Easting,E,Northing,N,AltMSL,NavStat,Hacc,Vacc,SOG,COG,Vvel,+ageC,HDOP,VDOP,TDOP,GU,RU,DR*hh<CR><LF>

Example:

\$PUBX,01,075142.00,467125.245,E,5236949.763,N,498.235,G3,2.1,1.9,0.005,85.63,0.0+00,,0.78,0.90,0.52,12,0,0*65

Only emitted by u-blox Antaris chipset.

PUBX 03 - u-blox Satellite Status

\$PUBX,03,GT{,ID,s,AZM,EL,SN,LK}*hh<CR><LF>

Example:

\$PUBX,03,11,23,-,,45,010,29,-,,46,013,07,-,,42,015,08,U,067,31,42,025,10,U,19+5,33,46,026,18,U,326,08,39,026,17,-,,32,015,26,U,306,66,48,025,27,U,073,10,36,+026,28,U,089,61,46,024,15,-,,39,014*0D

Only emitted by u-blox Antaris chipset.

(There's no PUBX 02)

PUBX 04 - u-blox Time of Day and Clock Information

\$PUBX,04,hhmmss.ss,ddmmyy,UTC_TOW,week,reserved,Clk_B,Clk_D,PG*hh<CR><LF>

Example:

\$PUBX,04,073731.00,091202,113851.00,1196,113851.00,1930035,-2660.664,43*3C<CR><+LF>

Only emitted by u-blox Antaris chipset.

TMVTD - Transas VTS / SML tracking system report

\$TMVTD,DDMMYY,hhmmss.ss,a,xxxx,c—c,IIII.IIII,a,yyyyy.yyyy,a,x.x,a,x.x,a,a*hh<CR><LF>

'TM' indicates message generated by SML tracking system. 'VTD' is name of the message.

Field Number:

1. Day/month/year (two-digit year, unknown base century)
2. Hour/minute/second to 0.1 sec precision, UTC.
3. 'R' indicates that this is an update for a radar track. No other values are valid
4. Internal unique ID number. Can't be changed even when target is re-identified
5. Target alias. Can be changed when the target identification data is edited.
Symbols "\$", "*", ",", and "." are not allowed to be used within the alias word. This field is variable length, at most 21 characters.
6. Latitude in degrees (two leading digits) and decimal minutes (trailing digits).
7. N or S for North or South latitude.
8. Longitude in degrees (three leading digits) and decimal minutes (trailing digits).
9. E or W for East or West longitude.
10. Target course in decimal degrees.
11. Fixed to T, indicates true course.
12. Target speed in decimal knots.
13. Fixed to N, indicates decimal speed in knots.
14. T or D. T = tracked, D = dropped. Message with status "dropped" is sent only once after target is dropped.
15. Checksum.

Transas is a manufacturer of proprietary ECDIS systems.

Described in [\[MALTESE\]](#), actually a Maltese government document.